

Julien Martinelli

Research Fellow, Finnish Center for Artificial Intelligence, Aalto University

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Appointments

Apr 2025-present	Research Fellow Finnish Center for Artificial Intelligence, Aalto University, Finland
Mar 2023-Mar 2025	Postdoctoral researcher Inria Bordeaux Sud-Ouest, Bordeaux Population Health, France
Feb 2022-Feb 2024	Postdoctoral researcher Department of Computer Science, Aalto University, Finland
Oct-Nov 2023	Visiting researcher Inria Bordeaux Sud-Ouest, Bordeaux Population Health, France
Jul-Aug 2021	Summer intern Centre International de Rencontres Mathématiques, France

Education

February 2022	PhD École polytechnique, Inria Saclay, Institut Curie Saint-Cloud, France Thesis: On learning mechanistic models from time series data with applications to personalised chronotherapies
September 2018	Master of Science in Applied Mathematics Université Paris 7 (now Université Paris-Cité), France Thesis: mechanistic model learning from time-series data
May 2015	Bachelor of Mathematics Université Paris 5 (now Université Paris-Cité), France

Supervision & Leadership

PhD advising with Prof. Harri Lähdesmäki and Prof. Samuel Kaski at Aalto University

04/2026-present	Tuan Vu
09/2025-present	Nicolas Blumer , <i>Preliminary title: Sequential Decision-making with Expert feedback.</i>
01/2025-present	Xinyu Zhang , <i>In-context Black-Box Optimization.</i>
06/2023-present	Marshal Sinaga , <i>Robustness and Human-Centric Aspects in Bayesian Optimization.</i>

Other supervision	(M) Master thesis (R) Research assistant (I) 2026 Summer intern (B) Bachelor thesis
06/2026–09/2026	(I) Mahtab Hosseinidolama, Mechanistic interpretability for tabular foundation models. (I) Tomas Remis, Gaussian processes for biological dynamical systems. (I) Quan Tran, High dimensional Bayesian optimization.
01/2026–present	(M) Barnik Roy Brata, In-context black-box optimization.
09/2025–03/2026	(M) Tuan Vu, Time-varying latent space Bayesian optimization.
06/2023–09/2024	(M) Xinyu Zhang, Preferential Amortized Black-Box Optimization. Also (R) (2023) Probabilistic Neural Additive Models.
06/2023–09/2023	(B) Duong Le, Benchmarking Bayesian experimental design for drug design.
06/2023–09/2023	(B) Kaul Rajat, Learning biological ODE models from time-series data.

Teaching & Academic Service

Teaching

- Fall 2025 **Modeling Biological Networks D**, Aalto University (MSc. level)
Gave 3 hours of lectures on data-driven inference of mechanistic ODEs and created the material myself.
- Involved as a teaching assistant in exercise sessions:
- **Advanced Probabilistic Methods**, Aalto University (MSc. level): Fall semester 2022
 - **Real Analysis**, Université Paris Cité (BSc. level): Spring semester 2020 and 2021
 - **Mathematics and Calculus**, Université Paris Cité (BSc. level): Spring semester 2019

Academic service

Reviewer: NeurIPS, ICML, ICLR, AISTATS, UAI (annually since 2024); AAAI, JMLR, PAMI, TCBB, ECCB, CMSB.

Membership: European Laboratory for Learning and Intelligent Systems Society (ELLIS).

Education and workforce development: Contributor to NSF Thrust 4 (EWD), supporting training material on bio-foundry workflows, and AI-enabled automation.

Invited talks

- European Conference on Mathematical and Theoretical Biology, Graz (2026), mini symposium “Automated discovery of dynamical digital twins from time series”
- European Conference on Mathematical and Theoretical Biology, Toledo (2024), mini symposium “Gaussian processes and inference for dynamical systems”
- Biostatistics Seminar, Bordeaux Population Health, France (2023)
- Public Seminar, CRiStAL team, Université de Lille, France (2023)
- AI Day, Finland (2022)
- Workshop on Hybrid models and methods in systems medicine, Institut Curie, Paris (2022)

Publication list

Publications

- X. Zhang, C. Hassan, **J. M.**, D. Huang, and S. Kaski. In-Context Multi-Objective Optimization. In *International Conference on Learning Representations*, 2026. [link](#)
- C. Métayer, A. Ballesta, and **J. M.** Data-driven Discovery of Digital Twins in Biomedical Research. *Briefings in Bioinformatics*, 2026. [link](#)
- J. M.** Position: Biology is the Challenge Physics-Informed ML needs to Evolve. In *Advances in Neural Information Processing Systems*, 2025. **6% acceptance rate**. [link](#)
- M. Sinaga, **J. M.**, and S. Kaski. Robust and Computation-Aware Gaussian Processes. In *Advances in Neural Information Processing Systems*, 2025. [link](#)
- S. Sloman, **J. M.**, and S. Kaski. Proxy-informed Bayesian Transfer Learning with Unknown Sources. In *Conference on Uncertainty in Artificial Intelligence*, 2025. [link](#)
- X. Zhang, D. Huang, S. Kaski, and **J. M.** PABBO: Preferential Amortized Black-Box Optimization. In *International Conference on Learning Representations*, 2025. **Spotlight (top 5%)**. [link](#)
- Y. Nahal, J. Menke, **J. M.**, M. Heinonen, M. Kabeshov, J. J. Paul, E. Nittinger, O. Engkvist, and S. Kaski. Human-in-the-loop Active Learning for Goal-Oriented Molecule Generation. *Journal of Cheminformatics*, 2024. [link](#)
- S. Sloman, A. Bharti, **J. M.**, and S. Kaski. Bayesian Active Learning in the Presence of Nuisance Parameters. In *Conference on Uncertainty in Artificial Intelligence*, 2024. **Oral**. [link](#)
- J. M.**, A. Bharti, A. Tiihonen, S. T. John, L. Filstroff, S. Sloman, P. Rinke, and S. Kaski. Learning Relevant Contextual Variables within Bayesian Optimization. In *Conference on Uncertainty in Artificial Intelligence*, 2024. [link](#)
- C. Frioux, S. Huet, S. Labarthe, **J. M.**, T. Malou, D. Sherman, M.-L. Taupin, and P. Ugalde-Salas. Accelerating Metabolic Models Evaluation with Statistical Metamodels: Application to Salmonella Infection Models. *ESAIM: Proceedings and Surveys*, 2023. [link](#)
- P. Mikkola, **J. M.**, L. Filstroff, and S. Kaski. Multi-Fidelity Bayesian Optimization with Unreliable Information Sources. In *International Conference on Artificial Intelligence and Statistics*, 2023. [link](#)
- J. Hesse*, **J. M.***, O. Aboumanify, A. Ballesta, and A. Relógio. A Mathematical Model of the Circadian Clock and Drug Pharmacology to Optimize Irinotecan Administration Timing in Colorectal Cancer. *Computational and Structural Biotechnology Journal*, 2021. Co-first author. [link](#)
- J. M.**, S. Dulong, X.-M. Li, M. Teboul, S. Soliman, F. Lévi, F. Fages, and A. Ballesta. Model Learning to Identify Systemic Regulators of the Peripheral Circadian Clock. *Bioinformatics*, 2021. [link](#)
- J. M.**, J. Grignard, S. Soliman, and F. Fages. On Inferring Reactions from Data Time Series by a Statistical Learning Greedy Heuristics. In *International Conference on Computational Methods in Systems Biology*, 2019a. [link](#)

Preprints

- N. Blumer, **J. M.**, and S. Kaski. In-Context Black-Box Optimization with Unreliable Feedback, 2026. arXiv. [link](#)
- J. M.**, I. Rebai, D. W. Haas, and J. Bertrand. Joint Bayesian Inference of Genetic Effect Sizes and PK Parameters in Nonlinear Mixed-Effects Models, 2026. arXiv. [link](#)
- T. A. Vu, **J. M.**, and H. Lähdesmäki. Time-Aware Latent Space Bayesian Optimization, 2026. arXiv. [link](#)
- M. Sinaga, **J. M.**, V. Garg, and S. Kaski. Anchor-Based Heteroscedastic Noise for Preferential Bayesian Optimization, 2024a. arXiv. [link](#)
- J. M.**, J. Grignard, S. Soliman, A. Ballesta, and F. Fages. Reactmine: a Statistical Search Algorithm for Inferring Chemical Reactions from Time Series Data, 2022. arXiv. [link](#)

Workshop communications

M. Sinaga, **J. M.**, and S. Kaski. Computation-Aware Robust Gaussian Processes. In *NeurIPS 2024 Workshop on Bayesian Decision-making and Uncertainty*, 2024b. [link](#)

X. Zhang, **J. M.**, and S. T. John. Challenges in Interpretability of Additive Models. In *IJCAI 2024 Workshop on Explainable Artificial Intelligence*, 2024. [link](#)

J. M., Y. Nahal, D. Lê, O. Engkvist, and S. Kaski. Leveraging Expert Feedback to Align Proxy and Ground Truth Rewards in Goal-Oriented Molecular Generation. In *NeurIPS 2023 Workshop on New Frontiers of AI for Drug Discovery and Development*, 2023b. [link](#)

J. M., A. Bharti, A. Tiihonen, L. Filstroff, S. T. John, S. Sloman, P. Rinke, and S. Kaski. Learning Relevant Contextual Variables within Bayesian Optimization. In *NeurIPS 2023 Workshop on Adaptive Experimental Design and Active Learning in the Real World*, 2023a. [link](#)

M. Sinaga, **J. M.**, and S. Kaski. Preferential Heteroscedastic Bayesian Optimization with Informative Noise Priors. In *NeurIPS 2023 Workshop on Adaptive Experimental Design and Active Learning in the Real World*, 2023. [link](#)

J. M., J. Grignard, S. Soliman, and F. Fages. A Statistical Unsupervised Learning Algorithm for Inferring Reaction Networks from Time Series Data. In *ICML 2019 Workshop on Computational Biology*, 2019b. [link](#)

PhD thesis

J. M. *On Learning Mechanistic Models from Time Series Data with Applications to Personalised Chronotherapies*. PhD thesis, École Polytechnique, 2022. [link](#)